

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 5**Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
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Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A uniform thin rod of mass $M = 2$ kg and length $L = 1.5$ m is pivoted about one end and allowed to swing as a physical pendulum. Given that the moment of inertia about one end is $I = \frac{1}{3}ML^2$ and the distance from the pivot to the centre of mass is $h = L/2$, what is the period of small oscillations? (Take $g = 9.81$ m/s².) [2 marks]

- A. 1.27 s
- B. 2.01 s
- C. 2.76 s
- D. 3.14 s

2. A particle undergoes simple harmonic motion with amplitude A and maximum speed $v_{\max}$. At what displacement from the equilibrium position is the speed of the particle equal to $v_{\max}/2$? [2 marks]

- A. $x = A/2$
- B. $x = A/4$
- C. $x = A\sqrt{3}/2$
- D. $x = A/\sqrt{2}$

3. An overdamped harmonic oscillator is displaced from equilibrium and released from rest. Which of the following best describes its subsequent motion? [2 marks]

- A. It oscillates with a slowly decreasing amplitude.
- B. It returns to equilibrium as quickly as possible without oscillating.
- C. It returns to equilibrium without oscillating, but more slowly than the critically damped case.
- D. It oscillates with constant amplitude indefinitely.

4. A forced oscillator has natural frequency $\omega_0 = 10$ rad/s and damping parameter $\gamma = 2$ s⁻¹. It is driven at angular frequency $\omega = 10$ rad/s (i.e. exactly at ω_0). What is the ratio $A(\omega)/A(0)$ of the steady-state amplitude at this driving frequency to the amplitude at zero driving frequency? [2 marks]

- A. 1
- B. 5
- C. 10
- D. 20

5. A transverse wave on a string has wavelength $\lambda = 0.50$ m and frequency $f = 120$ Hz. What is the wave speed? [2 marks]

- A. 24 m/s
- B. 60 m/s
- C. 240 m/s
- D. 600 m/s

6. A harmonic wave on a string has amplitude $A = 0.04$ m, angular frequency $\omega = 50$ rad/s, and travels with speed $v = 10$ m/s on a string of linear mass density $\mu = 0.05$ kg/m. What is the average power transmitted by the wave? [2 marks]

- A. 0.05 W
- B. 0.10 W
- C. 0.50 W
- D. 1.00 W

7. A single loudspeaker produces a sound level of 80 dB at a certain point. A second identical loudspeaker is placed next to the first, also producing the same intensity at that point. What is the combined sound level at that point? [2 marks]

- A. 80 dB
- B. 83 dB
- C. 86 dB
- D. 160 dB

8. A stationary siren emits sound at frequency $f = 500$ Hz. An observer moves directly away from the siren at speed $v_o = 34$ m/s. Taking the speed of sound as $v = 340$ m/s, what frequency does the observer hear? [2 marks]

- A. 450 Hz
- B. 455 Hz
- C. 500 Hz
- D. 550 Hz

9. A string fixed at both ends has a fundamental frequency of $f_1 = 150$ Hz. What is the frequency of the 5th harmonic? [2 marks]

- A. 30 Hz
- B. 300 Hz
- C. 600 Hz

D. 750 Hz

10. Two travelling waves of the same amplitude A and the same frequency f propagate in the same direction along a string. They have a constant phase difference of π radians. What is the amplitude of the resultant wave? [2 marks]

A. 0

B. A

C. $\sqrt{2}A$

D. $2A$

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. In Dirac notation, the bra $\langle\psi|$ lives in which mathematical space? [2 marks]
- A. The Hilbert space (the same space as kets)
 - B. The dual space of the Hilbert space
 - C. The space of linear operators on the Hilbert space
 - D. The space of eigenvalues of Hermitian operators
2. A quantum state is expanded in an orthonormal energy eigenbasis as $|\psi\rangle = \sum_n c_n |E_n\rangle$. The condition $\sum_n |c_n|^2 = 1$ is guaranteed at all times by which property of the time-evolution operator? [2 marks]
- A. Hermiticity
 - B. Linearity
 - C. Unitarity
 - D. Commutativity with the Hamiltonian
3. Given two orthonormal bases $\{|a_i\rangle\}$ and $\{|b_j\rangle\}$ for a Hilbert space, the matrix element for changing from the $\{|b_j\rangle\}$ basis to the $\{|a_i\rangle\}$ basis is obtained by inserting the identity. Which expression gives the components of $|\psi\rangle$ in the $\{|a_i\rangle\}$ basis? [2 marks]
- A. $\langle b_j|\psi\rangle$
 - B. $\langle a_i|\psi\rangle = \sum_j \langle a_i|b_j\rangle \langle b_j|\psi\rangle$
 - C. $|a_i\rangle\langle a_i| |\psi\rangle$
 - D. $\langle\psi|a_i\rangle$
4. What are the eigenvalues of the Pauli matrix $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$? [2 marks]
- A. 0 and 1
 - B. +1 and -1
 - C. $+\hbar/2$ and $-\hbar/2$
 - D. $+\hbar$ and $-\hbar$
5. A spin-1/2 particle is in the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$. What is the expectation value $\langle\sigma_z\rangle$ in this state? [2 marks]

- A. +1
- B. -1
- C. 0
- D. 1/2

6. A spin-1/2 particle is prepared in the state $|\uparrow_y\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + i|\downarrow_z\rangle)$. If S_z is measured, what are the probabilities of obtaining $+\hbar/2$ and $-\hbar/2$? [2 marks]

- A. $P(+\hbar/2) = 1, P(-\hbar/2) = 0$
- B. $P(+\hbar/2) = 0, P(-\hbar/2) = 1$
- C. $P(+\hbar/2) = 1/2, P(-\hbar/2) = 1/2$
- D. $P(+\hbar/2) = 1/4, P(-\hbar/2) = 3/4$

7. A two-level quantum system has energy levels E_1 and E_2 with $E_2 > E_1$. The system is prepared in a superposition of both energy eigenstates. The probability of finding the system in either eigenstate oscillates in time. What is the period of this oscillation? [2 marks]

- A. $T = 2\pi\hbar/E_1$
- B. $T = 2\pi\hbar/E_2$
- C. $T = 2\pi\hbar/(E_2 - E_1)$
- D. $T = 2\pi\hbar/(E_1 + E_2)$

8. The SWAP operator acts on a two-particle state by exchanging the states of the two particles. What is SWAP $|\uparrow\rangle_1 |\downarrow\rangle_2$? [2 marks]

- A. $|\uparrow\rangle_1 |\downarrow\rangle_2$
- B. $|\downarrow\rangle_1 |\uparrow\rangle_2$
- C. $-|\downarrow\rangle_1 |\uparrow\rangle_2$
- D. $\frac{1}{\sqrt{2}}(|\uparrow\rangle_1 |\downarrow\rangle_2 + |\downarrow\rangle_1 |\uparrow\rangle_2)$

9. The wave function of a system of two identical bosons must satisfy which symmetry requirement under particle exchange? [2 marks]

- A. It must be antisymmetric: $\text{SWAP} |\Psi\rangle = -|\Psi\rangle$
- B. It must be symmetric: $\text{SWAP} |\Psi\rangle = +|\Psi\rangle$
- C. It can be either symmetric or antisymmetric
- D. It must vanish: $\text{SWAP} |\Psi\rangle = 0$

10. In the Standard Model, what is the electric charge of the up quark? [2 marks]

- A. $-1/3 e$
- B. $+1/3 e$
- C. $+2/3 e$
- D. $+1 e$

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

Rocket Equation, Impulse, and Escape Velocity

- (a) A single-stage rocket has initial mass $m_0 = 5000$ kg and exhaust velocity $u = 3000$ m/s relative to the rocket. After burning all its fuel, the final mass is $m_f = 1000$ kg. Using the Tsiolkovsky rocket equation, find the maximum speed change Δv . If the rocket starts from rest in deep space (no gravity), what is its final speed? 4
- (b) During the burn, the rocket engine exerts a thrust of $F_{\text{thrust}} = 15,000$ N. Calculate the mass flow rate of the exhaust $|dm/dt|$. Find the initial acceleration a_0 of the rocket in deep space. 3
- (c) Calculate the escape velocity from the surface of the Earth. If a projectile is launched vertically at 8.0 km/s from the Earth's surface, find the maximum height it reaches above the surface. (Use $M_{\oplus} = 5.97 \times 10^{24}$ kg, $R_{\oplus} = 6.37 \times 10^6$ m, $G = 6.674 \times 10^{-11}$ N·m²/kg².) 4
- (d) A ball of mass 0.05 kg travelling at 20 m/s hits a wall and bounces back at 15 m/s. The contact time is 0.01 s. Find the impulse delivered to the ball and the average force exerted by the wall on the ball. 4

Question 2.

[15 marks]

Heat Engines and Entropy

- (a) State the first and second laws of thermodynamics. Define entropy in terms of a reversible heat transfer. 3
- (b) A Carnot engine operates between a hot reservoir at $T_h = 800$ K and a cold reservoir at $T_c = 200$ K. It absorbs $Q_h = 4000$ J per cycle from the hot reservoir. Find the efficiency of the engine, the work done per cycle, and the heat rejected to the cold reservoir per cycle. 4
- (c) A block of aluminium of mass 5 kg and specific heat capacity $c = 900$ J/(kg·K) at an initial temperature of 600 K is placed in a large lake at 300 K (which acts as an infinite thermal reservoir). Find: the final equilibrium temperature, the entropy change of the aluminium, the entropy change of the lake, and the total entropy change of the universe. Verify that the second law is satisfied. 4
- (d) Two identical blocks of metal, each of mass 1 kg and specific heat capacity $c = 500$ J/(kg·K), are initially at temperatures $T_1 = 400$ K and $T_2 = 200$ K. They are brought into thermal contact and allowed to reach equilibrium. Find the final temperature. Calculate the total entropy change. If instead a reversible (Carnot) engine were operated between the two blocks until they reached the same final temperature, what is the maximum work that could be extracted? 4

Question 3.

[15 marks]

Wave Reflection, Standing Waves, and Energy

- (a) A transverse wave is incident on a boundary between two strings with impedances Z_1 and Z_2 . Using the boundary conditions of continuity of displacement and continuity of the transverse force at the junction, derive the reflection and transmission amplitude coefficients:

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}.$$

3

- (b) A wave $y_i = 0.03 \sin(5x - 40t)$ m travels on string 1, which has linear mass density $\mu_1 = 0.02$ kg/m, towards a junction with string 2, which has linear mass density $\mu_2 = 0.08$ kg/m. Both strings are under the same tension. Calculate the tension in the strings, the wave speeds on each string, and the impedances Z_1 and Z_2 . 4
- (c) Find the amplitudes of the reflected and transmitted waves. Write the complete wave equations $y_r(x, t)$ and $y_t(x, t)$ for the reflected and transmitted waves. 4
- (d) Verify that the power reflection coefficient $R = r^2$ and power transmission coefficient $T_{\text{power}} = (Z_2/Z_1) t^2$ satisfy $R + T_{\text{power}} = 1$. Calculate the average power of the incident wave. 4

Question 4.

[15 marks]

Two-Level System and Interferometry

A quantum system has two energy levels $E_1 = 0$ and $E_2 = E_0$, with corresponding orthonormal eigenstates $|1\rangle$ and $|2\rangle$.

- (a) At time $t = 0$, the system is in the state $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle)$. Find the state $|\psi(t)\rangle$ at a later time t . Calculate the probability of finding the system in state $|1\rangle$ at time t . 4
- (b) Define the operator $\hat{P} = |1\rangle\langle 2| + |2\rangle\langle 1|$. Calculate the expectation value $\langle \hat{P} \rangle(t) = \langle \psi(t) | \hat{P} | \psi(t) \rangle$. 3
- (c) Consider a Mach–Zehnder interferometer with a phase shifter of angle ϕ in the top arm. The beam splitter acts as:

$$|T\rangle \rightarrow \frac{1}{\sqrt{2}}(|T\rangle + i|B\rangle), \quad |B\rangle \rightarrow \frac{1}{\sqrt{2}}(i|T\rangle + |B\rangle).$$

Each mirror gives a phase factor of i (i.e. $|T\rangle \rightarrow i|T\rangle$, $|B\rangle \rightarrow i|B\rangle$). Starting from input $|T\rangle$, trace the photon state through the interferometer (first beam splitter, mirrors, phase shifter on top arm, second beam splitter). Show that the output state is proportional to

$$i(e^{i\phi} - 1)|T\rangle + (-e^{i\phi} - 1)|B\rangle,$$

and hence find $P(T)$ and $P(B)$. 4

- (d) A “which-way” detector is placed in the top arm that records whether the photon took the top path, without destroying the photon. Explain qualitatively what happens to the interference pattern. Write the state of the combined photon+detector system after the first beam splitter and the detector interaction. Show that the detection probabilities at the output become $P(T) = P(B) = 1/2$, regardless of ϕ . 4

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v'_1 = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m